

4. Collisions		
4.1	Momentum as a vector. The impulse-momentum principle in vector form. Conservation of linear momentum.	
4.2	Direct impact of elastic particles. Newton's law of restitution. Loss of mechanical energy due to impact.	Students will be expected to know and use the inequalities $0 \leq e \leq 1$ (where e is the coefficient of restitution).
4.3	Successive impacts of up to three particles or two particles and a smooth plane surface.	Collision with a plane surface will not involve oblique impact.